1. I chose thirty-five samples because the Central Limit Theorem can be applied so that the distribution of sample means is closer to the normal distribution. Sampling method I use is simple random sampling, because this method is the most fair, there will be no bias.

2.

```
# 从Class A随机抽取30个样本
set.seed(123)  # 确保结果可重现
class_a_scores <- c(66,95,87,84,88,80,88,68,78,73,53,86,88,81,7
                    90,55,74,84,55,75,83,91,82,70,84,80,92,82,9
                    69,89,62,71,96,80,88,77,70,64,93,83,82,94,9
                    78,77,80,60,73,84,84,82,73,74,82,88,85,70,8
                    75,92,82,84,91,83,71,79,87,29,82,71,80,57,8

sample_scores <- sample(class_a_scores, 30)

# 描述性统计
cat("样本数据：\n")
print(sample_scores)
cat("\n样本均值：", mean(sample_scores), "\n")
cat("样本标准差：", sd(sample_scores), "\n")
cat("样本量：", length(sample_scores), "\n")

# 95%置信区间
conf_int_result <- t.test(sample_scores, conf.level = 0.95)
cat("\n95%置信区间结果：\n")
print(conf_int_result)
```

Output：

```
95%置信区间结果：
> print(conf_int_result)

        One Sample t-test

data:  sample_scores
t = 33.503, df = 29, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 72.70633 82.16034
sample estimates:
mean of x
 77.43333
```

We are 95 percent sure that the overall mean of Class A students 'scores is between 72.70633 and 82.16034. This means that approximately 95% of similarly constructed intervals will contain the true population mean if repeated multiple times.

3.

null hypothesis H: μ = 70 (population mean = 70 points)

```
# 假设检验
hypothesis_test <- t.test(sample_scores, mu = 70, alternative = "two.sided")
cat("\n假设检验结果：\n")
print(hypothesis_test)

# 提取关键值
cat("\n关键统计量：\n")
cat("t统计量：", hypothesis_test$statistic, "\n")
cat("p值：", hypothesis_test$p.value, "\n")
cat("自由度：", hypothesis_test$parameter, "\n")
```

Output：

```
> cat("t统计量：", hypothesis_test$statistic, "\n")
t统计量：  3.216176
> cat("p值：", hypothesis_test$p.value, "\n")
p值：  0.003184004
> cat("自由度：", hypothesis_test$parameter, "\n")
自由度：  29
```

Because p<0.05 so reject the null hypothesis.

4.

p-value = 0.003184004, which means that if the population mean is really equal to 70 points, the probability of observing the current sample mean or more extreme is almost zero.

The p-value measures the strength of the evidence against the null hypothesis, and the smaller the p-value, the stronger the evidence against the null hypothesis.

5.

If the sample size doubles (from 30 to 60), the confidence interval narrows. Specifically:

Current interval width: 82.16034 -72.70633= 9.45401 minutes

Expected new interval width: about 9.45401/√2 ≈ 6.68499 minutes

This is because the standard error (σ/√n) decreases with increasing sample size, making the confidence interval more precise.

6.

Using the t-distribution:

Small sample size (n 30)

Population standard deviation unknown

population obeys normal distribution or sample size is large enough (n > 30)

Use of the z distribution:

Large sample size ($n > 30$) and known population standard deviation

The population obeys a normal distribution and the population standard deviation is known

The reason for choosing the t-distribution for this experiment:

Although the sample size $n=30$ satisfies the condition of the central limit theorem, the population standard deviation is unknown and needs to be estimated by the sample standard deviation, so it is more appropriate to use the t-distribution. The t distribution is more conservative than the z distribution and provides more accurate inferences when sample sizes are small.